

E1 — Embedded Track, Week 1

Member: _____

Project setup (do first)

1. Fork the repo: <https://github.com/slvDev/esp32-ai>
2. Clone your fork. Add the original as upstream.
3. Read README.md, RESULTS.md, and research/tinystories/README.md.
4. Write 5 lines in the shared Drive doc: what you think the project does and where the hard part is.

Your task this week

Goal: the toolchain works and the firmware sources are mapped.

Steps: 1. Install the ESP32 toolchain. Use Arduino ESP32 core 3.3.10 or ESP-IDF. - Arduino ESP32 core: <https://github.com/espressif/arduino-esp32> - ESP-IDF: <https://github.com/espressif/esp-idf> 2. Build the portable C runtime on host. It must compile without a board. 3. Read the firmware folders: esp32_barista, esp32_tinystories, benchmarks. 4. Write the firmware walkthrough notes: what each folder does.

Deliverable: the walkthrough notes in the club repo.

Verify: the host runtime builds without error. You can name the parts of a firmware sketch.

Learning this week

The ESP32-S3 memory map. Draw it: 512 KB SRAM, 8 MB PSRAM, 16 MB flash. Label what the model puts in each tier.

Resources

- ESP32-S3 docs: <https://docs.espressif.com/projects/esp-idf/en/latest/esp32s3/>
- firmware/: <https://github.com/slvDev/esp32-ai/tree/main/firmware>

Done when

- ☐ Toolchain installed
- ☐ Host runtime builds without error
- ☐ Walkthrough notes exist in the club repo
- ☐ You can say what the project does in one sentence